Protein Sequence Determination

Determining the primary structure (amino acid sequence) involves 3 main stages:

1. Determination of amino acid composition.
2. Degradation (cleavage) into smaller fragments.
3. Determination of the exact sequence.

1. Amino Acid Composition

The protein is completely hydrolyzed to liberate individual amino acids using:

- Acid Hydrolysis: e.g., HCl (6M, 110°C).
- Alkaline Hydrolysis: e.g., $Ba(OH)_2$.
- Enzymatic Hydrolysis: e.g., Pronase (a mixture of non-specific enzymes).

Amino acids are then separated and identified (e.g., via Ion-exchange chromatography).

2. Protein Denaturation & Folding

Before sequencing, higher-order structures must be disrupted (Denaturation) without breaking peptide bonds.

- Denaturing Agents: Heat, extreme pH, SDS (detergent), Urea (6M - disrupts H-bonds).
- Reducing Agents: 2-Mercaptoethanol, Dithiothreitol (DTT), Performic acid (break Disulfide -S-S- bonds).

> Anfinsen Experiment (1957): Used RNase A denatured with urea and 2-mercaptoethanol. When reagents were removed, it refolded and regained activity. Conclusion: A protein's primary structure dictates its 3D structure!

Molecular Chaperones (Heat Shock Proteins): Assist in correct protein folding in cells. Prevent toxic misfolding, which leads to aggregates (amyloid plaques) seen in Alzheimer's, Parkinson's, and Prion diseases.

3. Cleavage of Polypeptide Chains

Large chains are chopped into manageable fragments using specific reagents:

Reagent / Enzyme	Specificity (Where it cleaves)
Edman's Reagent	Removes N-terminal amino acid
Cyanogen Bromide	Hydrolyzes C-side of Methionine (Met)
Carboxypeptidase A	Removes C-terminal AA (Except Arg, Lys)
Carboxypeptidase B	Removes C-terminal AA (ONLY Arg, Lys)
Trypsin	Hydrolyzes C-side of Arg & Lys (Basic AAs)
Chymotrypsin	Hydrolyzes C-side of Phe, Tyr, Trp (Aromatics)
Elastase	Hydrolyzes C-side of Gly, Ala (Small AAs)

4. Sequencing Methods

- Sanger's Method: Uses FDNB (1-fluoro-2,4-dinitrobenzene). Binds to N-terminal AA to form a DNP-derivative. Only identifies the very first N-terminal AA.
- Edman's Degradation: Uses Phenyl isothiocyanate. Reacts with N-terminal AA to release a PTH-amino acid. Advantage: Allows sequential labeling and removal of AAs one by one without destroying the whole chain!
- Sequenator: An automated machine based on Edman's degradation (takes ~2 hrs per AA).
- Reverse Sequencing: Deducing the AA sequence from the DNA sequence. Drawback: Cannot identify disulfide bonds or post-translational modifications.

Overlapping Peptides: By using different cleavage methods (e.g., Trypsin then Cyanogen bromide) simultaneously, we create overlapping fragments. Comparing these overlaps allows us to map the full intact sequence!